

CHAPTER 14

The Somatic Nervous System

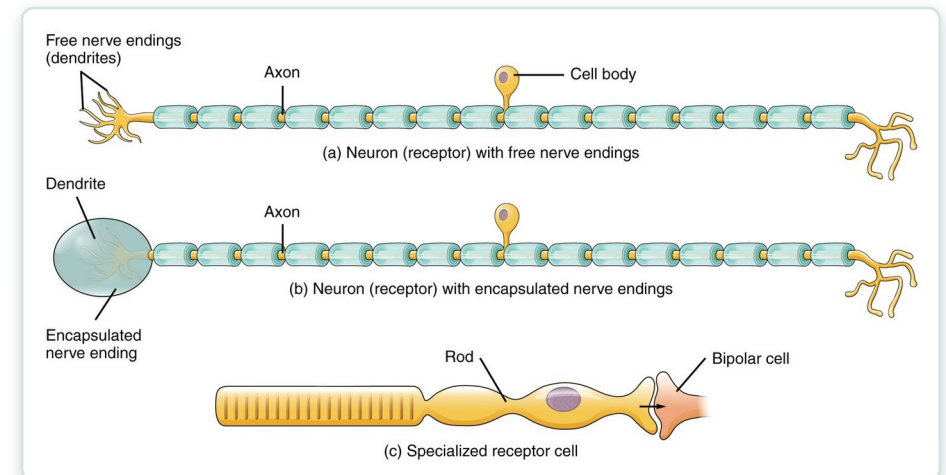
Sensation, central processing, and voluntary movement.



Learning Objectives

After studying this chapter, students will be able to:

- 1 Describe the components of the somatic nervous system
- 2 Name the modalities and submodalities of the sensory systems
- 3 Distinguish between general and special senses
- 4 Describe regions of the central nervous system that contribute to somatic functions



Sensory receptors detect stimuli from the world and the body.



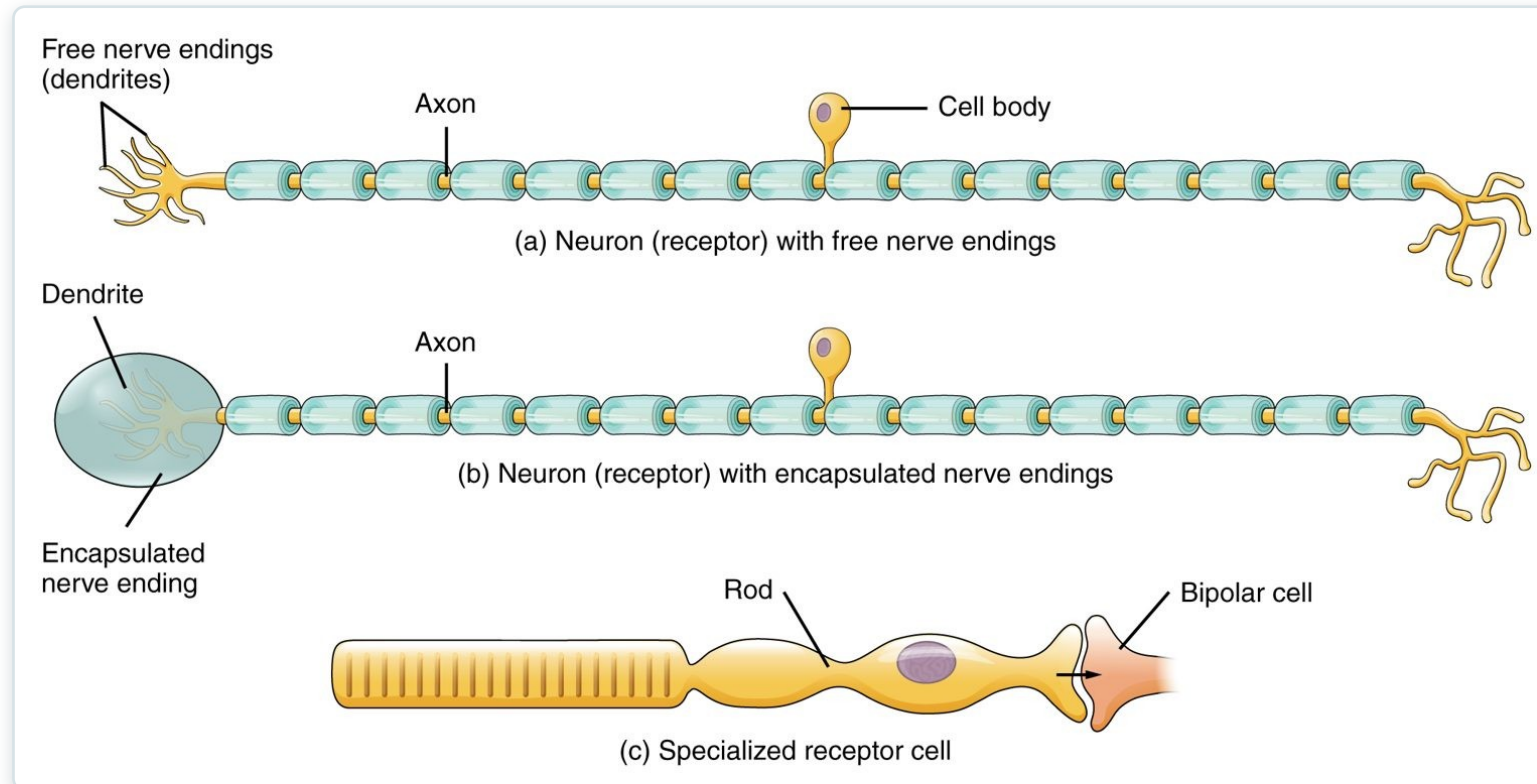
14.1

SECTION

Sensory Perception

How receptors detect stimuli — the general and special senses.

Sensory Receptors



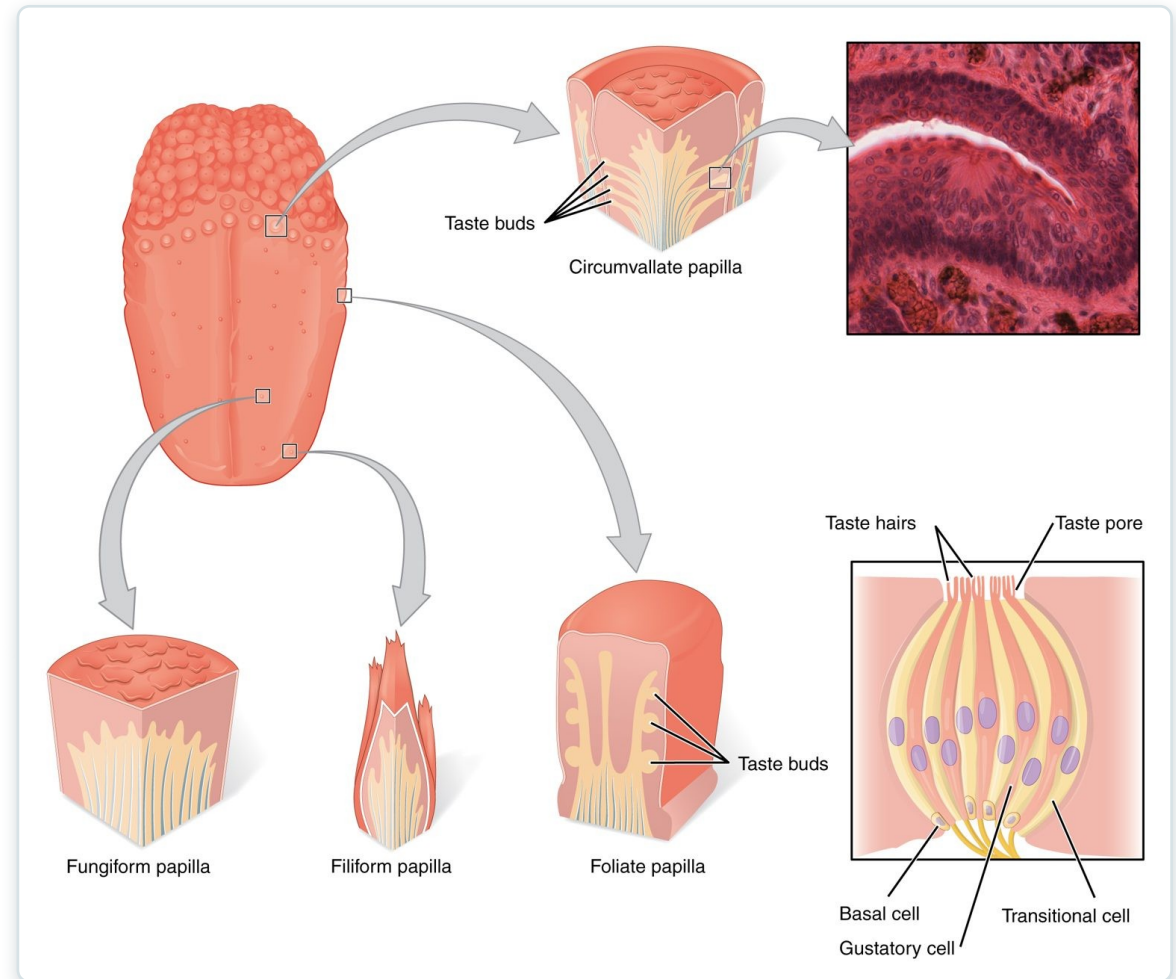
Types of sensory receptors

- Receptors detect stimuli
- Convert them to nerve signals
- Respond to touch, temperature, pain
- Free and encapsulated endings

SECTION 14.1

Taste (Gustation)

- Taste buds detect dissolved chemicals
- Located on tongue papillae
- Five basic tastes
- A chemical (chemoreceptor) sense

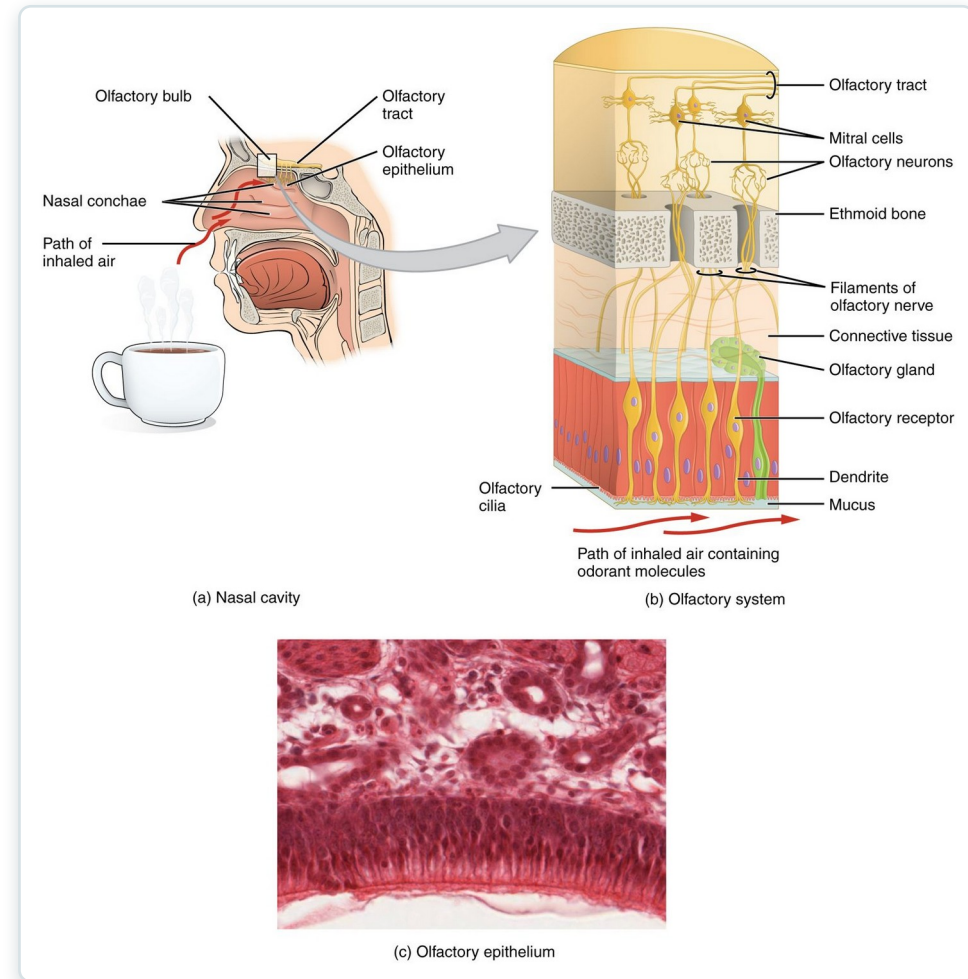


Taste buds and the sense of taste

SECTION 14.1

Smell (Olfaction)

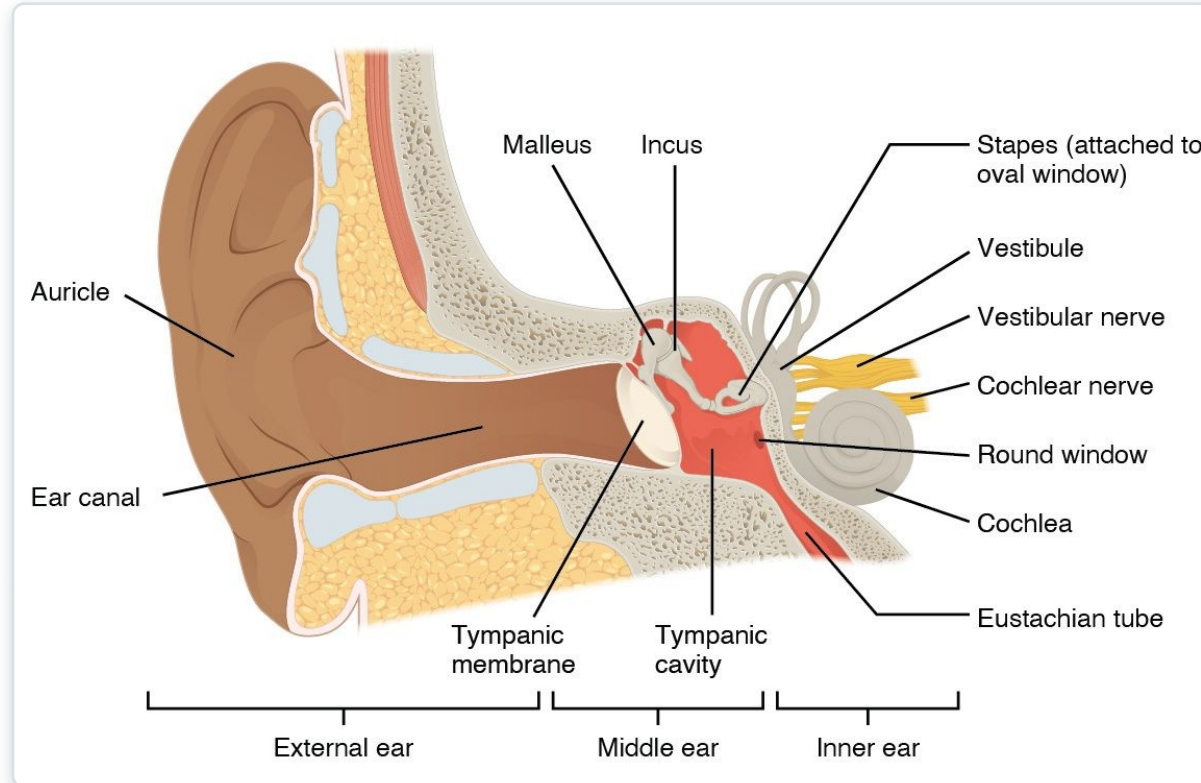
- Olfactory receptors in the nasal cavity
- Detect airborne chemicals
- Signal travels to the brain directly
- Strongly tied to memory



The olfactory system

SECTION 14.1

Hearing — The Ear

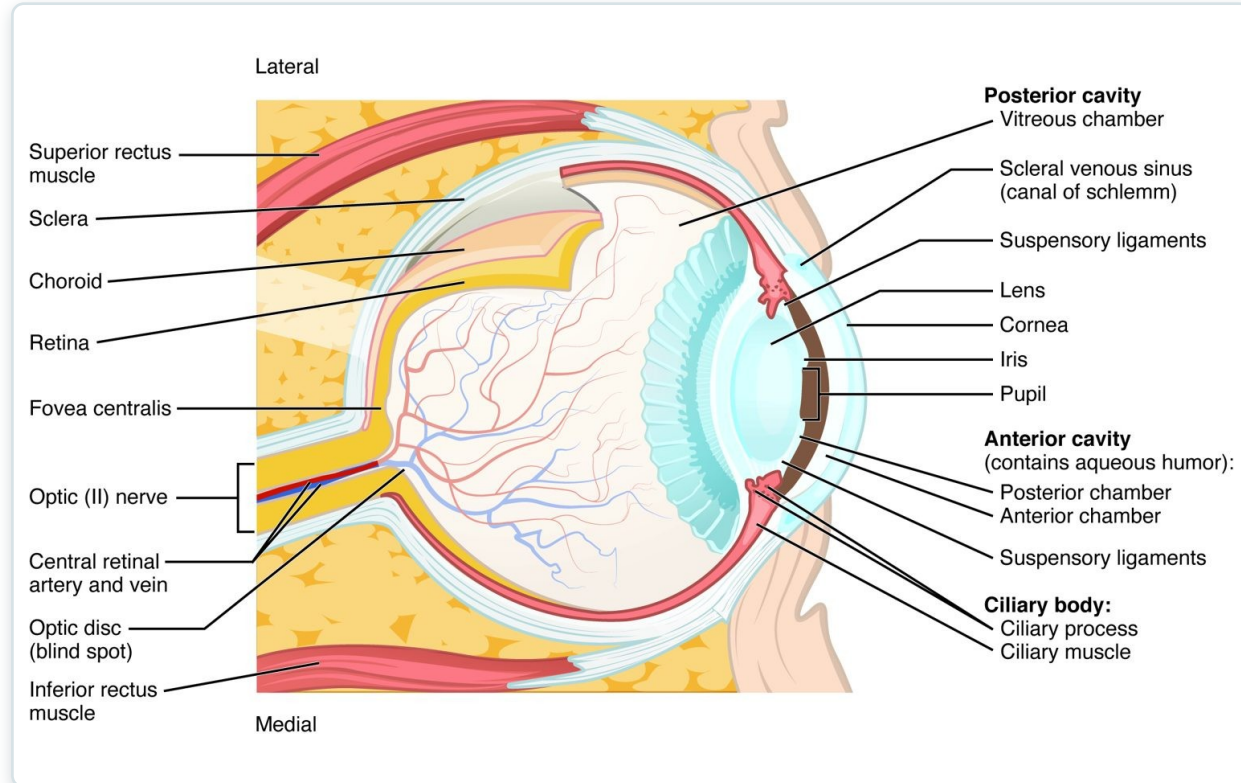


The anatomy of the ear

- Outer ear gathers sound
- Middle ear bones amplify it
- Inner ear cochlea transduces it
- Hair cells signal the brain

SECTION 14.1

Vision — The Eye



The anatomy of the eye

- The cornea and lens focus light
- The iris controls light entry
- The retina holds photoreceptors
- Rods and cones signal the brain



14.2

SECTION

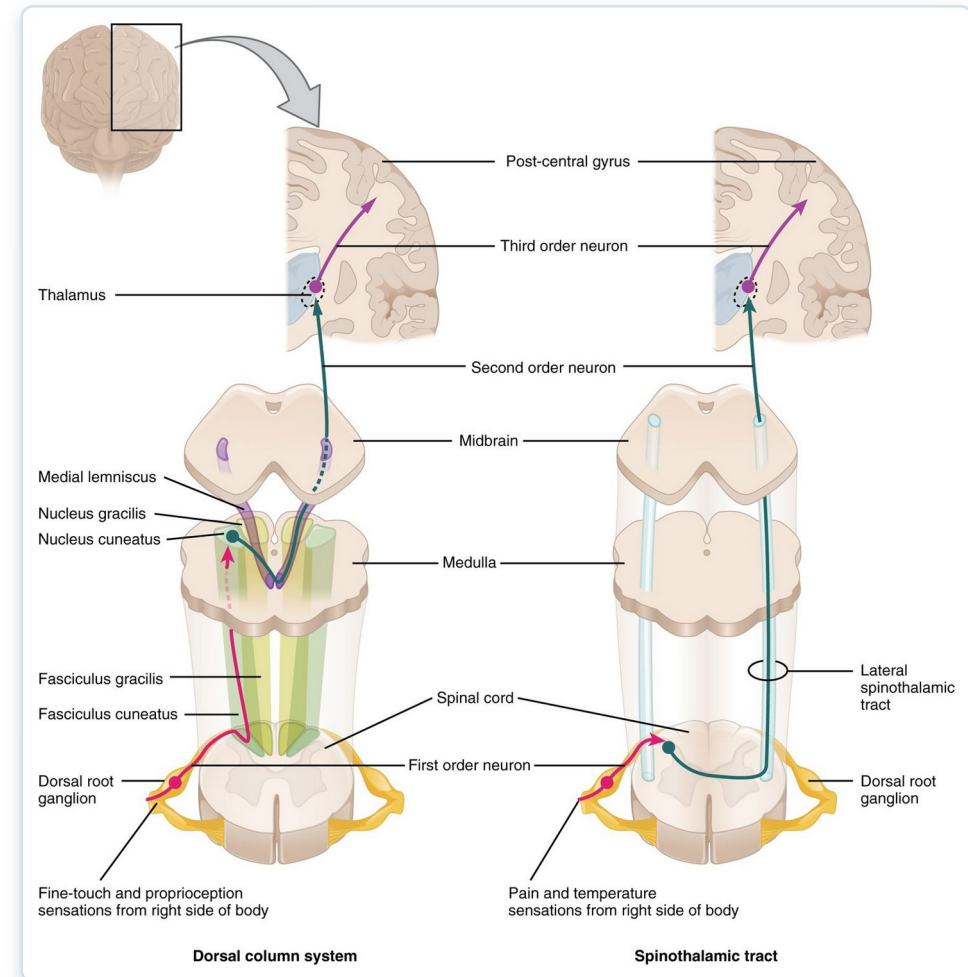
Central Processing

How the CNS processes sensory information into perception.

SECTION 14.2

Ascending Sensory Pathways

- Sensory signals travel to the brain
- Cross to the opposite side
- Relay through the thalamus
- Reach the sensory cortex

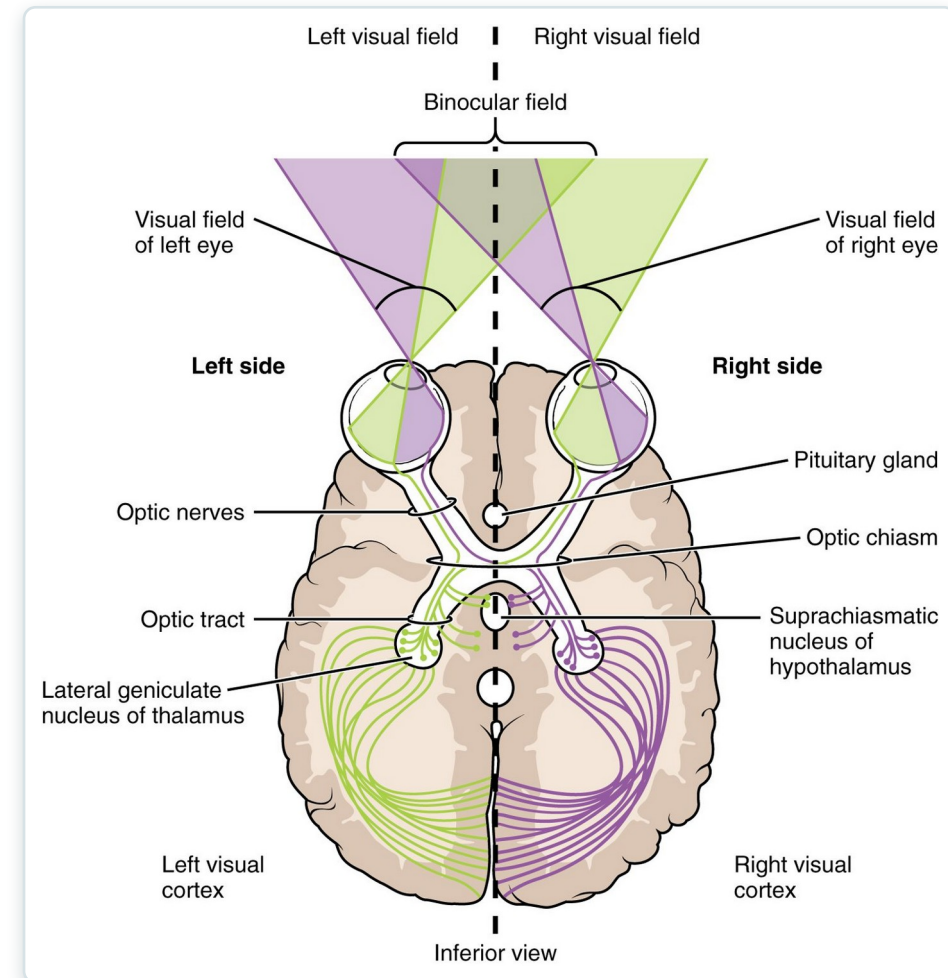


Ascending sensory pathways

SECTION 14.2

The Visual Pathway

- Optic nerves carry signals from each eye
- They meet at the optic chiasm
- Fibers partly cross
- End in the visual cortex

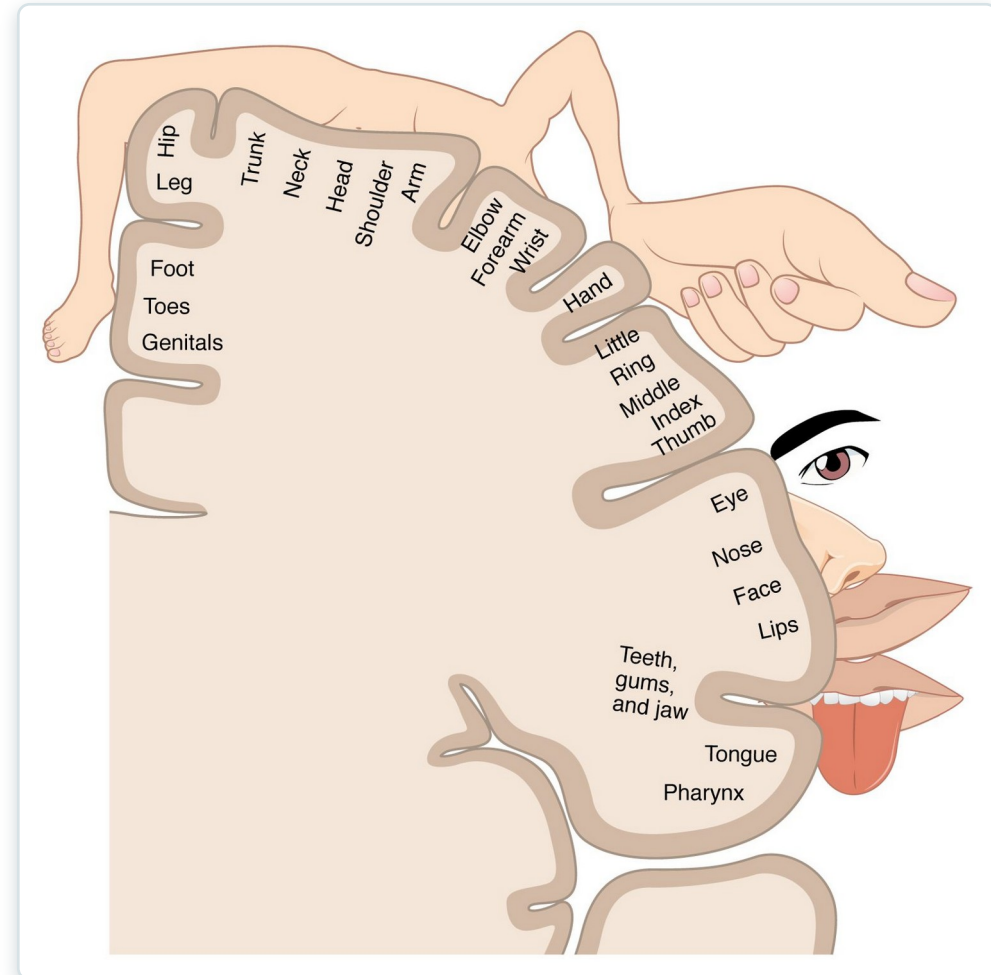


The visual pathway to the cortex

SECTION 14.2

The Sensory Homunculus

- The cortex maps the whole body
- Each part has a cortical area
- Sensitive areas get more space
- Hands and face dominate



The sensory homunculus



14.3

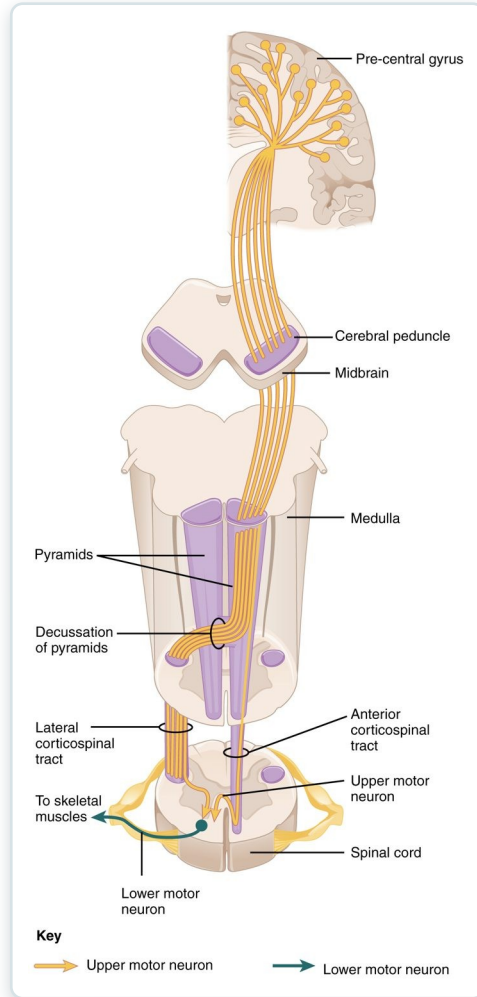
SECTION

Motor Responses

How the nervous system produces voluntary movement.

SECTION 14.3

The Motor Pathway



The descending motor pathway

- Commands start in the motor cortex
- Travel down the corticospinal tract
- Cross to the opposite side
- Upper and lower motor neurons

Key Takeaways

- Sensation begins when receptors transduce stimuli into nerve signals.
- The special senses — taste, smell, hearing, balance, and vision — use specialized receptors.
- Sensory pathways cross the midline and relay through the thalamus to the cortex.
- The cortex maps the body, devoting the most area to the most sensitive parts.
- Voluntary movement travels from the motor cortex down crossed corticospinal pathways.

CLINICAL CONNECTIONS

Why the Somatic Nervous System Matters in Practical Nursing

Sensory Loss

Numbness or deficits help localize nerve or CNS damage.

Hearing & Vision

Common deficits that affect safety and communication.

Pain Management

Understanding pain pathways guides effective relief.

Stroke Effects

Crossed pathways explain one-sided weakness or sensory loss.

Pupillary Reflex

Pupil response is a key neurological assessment.

Motor Deficits

Upper vs. lower motor neuron signs guide diagnosis.